

Neuroscience and Biobehavioral Reviews

Novel cognitive functions arise as macroscale gradients converge

--Manuscript Draft--

Manuscript Number:	NEUBIOREV-D-20-00398
Article Type:	Review Article
Keywords:	higher-order cognition; convergence; macroscale gradients; hierarchy; abstraction; nexus; TPJ
Corresponding Author:	Heejung Jung Dartmouth College UNITED STATES
First Author:	Heejung Jung
Order of Authors:	Heejung Jung
	Tor D. Wager
	R. McKell Carter
Abstract:	We summarize evidence for a convergence framework in which regions of the brain that support complex and novel cognitive functions are found at the convergence of sensory processing streams. Various described as convergence-divergence, associative, amodal, and polymodal, these regions of the brain have been shown to participate in a wide range of cognitive processes. However, there is little agreement about the specific role these regions play. Using the temporal-parietal junction (TPJ) as a template, we present a framework of converging macroscale functional gradients. As gradients travel from primary sensory cortices to higher-order brain regions, information becomes abstracted and hierarchical, and eventually meet at a point maximally and equally distant from their sensory origins. This convergence produces multifaceted combinations that result in new, emergent cognitive and affective faculties, which allow people to flexibly navigate a complex world.
Suggested Reviewers:	Joaquim Radua radua@clinic.cat
	Daniel S. Margulies daniel.margulies@icm-institute.org
	Julia M. Huntenberg huntenburg@cbs.mpg.de
	Rogier Mars rogier.mars@ndcn.ox.ac.uk



University of Colorado
Boulder

Institute of Cognitive Science
Muenzinger Psychology Building, Room D414
UCB 344
Boulder, Colorado 80309

t 303 735 2553
f 303 492 7177
mckell.carter@colorado.edu

August 6, 2020

Reference:

Dear Colleagues,

We submit our manuscript “Novel cognitive functions arise as macroscale gradients converge” for publication in *Neuroscience and Biobehavioral Reviews*. Convergence (or polymodal) regions of the brain have been shown to participate in a wide range of cognitive processes with little agreement about the specific role these regions play. Their function and the mechanisms by which they arise have been the focus of significant debate. Recent work has begun to show that these brain regions may not support any one of these functions but rather multifaceted combinations. Using one specific brain region, the temporal-parietal junction (TPJ), as a template, we present a framework of macroscale gradients in which convergence provides a rich setting for information to be integrated in a manner that gives rise to novel, multifaceted cognitive functions. We review multiple lines of research and come to the conclusion that higher-order representations can be synthesized and compared in unique ways where hierarchical sensory processing streams converge in the brain.

Our review is well suited for the scope of *Neuroscience and Biobehavioral Reviews*, it synthesizes a wide range of neuroscience perspectives from cognitive to structural and evolutionary properties of the brain. Please feel free to reach out with any further questions.

Sincerely,

A handwritten signature in black ink, appearing to read 'R. McKell Carter'.

R. McKell Carter, Ph.D.
Assistant Professor of Psychology and Neuroscience
University of Colorado Boulder

Highlights

- Function in higher-order brain regions is the source of extensive debate.
- Higher-order regions are furthest away, both anatomically and functionally, from sensory cortices.
- Hierarchical processing streams converge at points furthest from sensory cortices.
- Convergence brings highly abstracted representations into close proximity.
- Novel cognitive functions arise where abstracted representations can be combined.

Abstract

We summarize evidence for a convergence framework in which regions of the brain that support complex and novel cognitive functions are found at the convergence of sensory processing streams. Various descriptions of these regions include convergence-divergence, associative, amodal, and polymodal. These regions of the brain have been shown to participate in a wide range of cognitive processes. However, there is little agreement about the specific role these regions play. Using the temporal-parietal junction (TPJ) as a template, we present a framework of converging macroscale functional gradients. As gradients travel from primary sensory cortices to higher-order brain regions, information becomes abstracted and hierarchical, and eventually meet at a point maximally and equally distant from their sensory origins. This convergence produces multifaceted combinations that result in new, emergent cognitive and affective faculties, which allow people to flexibly navigate a complex world.

Title: Novel cognitive functions arise as macroscale gradients converge

Authors

Heejung Jung^{a,b,c}

Tor D. Wager^{a,c}

R. McKell Carter^{a,b,d*}

Affiliations

^a Institute of Cognitive Science, University of Colorado Boulder, Boulder, CO, United States

^b Department of Psychology and Neuroscience, University of Colorado Boulder, Boulder, CO,
United States

^c Department of Psychological and Brain Sciences, Dartmouth College, Hanover, NH, United
States

^d Department of Electrical, Computer & Energy Engineering, University of Colorado Boulder,
Boulder, CO, United States

*Please contact mckell.carter@colorado.edu for correspondence.

Abstract

We summarize evidence for a convergence framework in which regions of the brain that support complex and novel cognitive functions are found at the convergence of sensory processing streams. Variously described as convergence-divergence, associative, amodal, and polymodal, these regions of the brain have been shown to participate in a wide range of cognitive processes. However, there is little agreement about the specific role these regions play. Using the temporal-parietal junction (TPJ) as a template, we present a framework of converging macroscale functional gradients. As gradients travel from primary sensory cortices to higher-order brain regions, information becomes abstracted and hierarchical, and eventually meet at a point maximally and equally distant from their sensory origins. This convergence produces multifaceted combinations that result in new, emergent cognitive and affective faculties, which allow people to flexibly navigate a complex world.

Keywords

higher-order cognition; convergence; macroscale gradients; hierarchy; abstraction; nexus; TPJ

Main text

1. Introduction

Higher-order cognitive functions are important for humans to navigate a complex world. Our daily interactions require contributions from these functions, including attention, memory, value computation, cognitive control, language and social processes; all are computationally expensive processes that we carry out effortlessly, but are only beginning to understand. What we do know is that these functions are not allocated randomly in the brain. Some of these regions — including the temporal-parietal junction (TPJ), medial prefrontal cortex, and the posterior cingulate cortex — serve as hubs, commonly active in many tasks. While there are a number of theories on how these multi-sensory areas function (Man et al., 2013), the mechanisms by which higher-order cognitive functions (e.g. theory of mind) occur is yet to be explored.

One fundamental, but overlooked, aspect to understanding higher-order cognition is the role of lower-order sensory processing streams, or gradients. Sensory processing areas are, by definition, devoted to processing a single sensory modality. This area then provides inputs to produce the next, relatively more complex and abstracted representation in a neighboring cortex. Visual inputs, for example, are processed in the occipital lobe, beginning primarily with neurons that detect line orientations in very small areas of the visual field (Hubel & Wiesel, 1962). Moving further away from primary visual cortex, receptive fields grow larger and preferred stimuli become complex surfaces, textures, and eventually, objects (Fig. 1a). A similar structure is present for auditory (Fig. 1b) and somatosensory (Fig. 1c) cortices. Therefore, areas of the brain reaching from sensory areas to higher-order hubs can be viewed as hierarchical processing pipelines (Van Essen & Maunsell, 1983) anchored in primary sensory cortices and culminating

in polymodal and highly abstracted representations. This description of overall brain structure mirrors an emergence of diffusion measures of information processing in the brain (Larivière et al., 2020; Paquola et al., 2019; Vázquez-Rodríguez et al., 2019). Where processing streams converge, these areas combine the convergent information, and this novel mixture of information gives rise to multifaceted and unique functions, characteristic of higher-order cognition.

Using the TPJ as an example, we present a framework of convergence of macroscale gradients. We argue that convergence provides a rich setting for information to be integrated in a manner that gives rise to novel, multifaceted cognitive functions. We then focus on answering two questions: 1) how does convergence arise in the brain, and 2) what happens when processing streams converge? To answer these questions, we first offer a survey of the function in the TPJ, which will provide a guide for understanding multifaceted functions in convergent brain areas. We then summarize evidence for convergent processing streams and novel multifaceted functions in convergence zones.

2. TPJ

There are several regions of the brain that support higher-order cognition. Various called convergence-divergence (Damasio, 1989), associative (Buckner & Krienen, 2013), amodal (Fairhall & Caramazza, 2013; Tamber-Rosenau et al., 2013), transmodal (Mesulam, 1998), and polymodal (Baumgaertner et al., 2007), these brain regions have been shown to participate in a wide range of cognitive processes with little agreement about the specific role each region plays. The TPJ is such a region (Bukowski et al., 2017). We review the debate over function in the TPJ, starting with single modular views, then describing a domain-specific view in which single functions split the TPJ, and finally moving toward a more domain-general view.

Early work characterizing function in the TPJ was a series of contributions from different areas of cognitive neuroscience, each claiming that the TPJ was involved only in attention, memory, or social processes. Evidence that the TPJ was an attention module was based on neuroimaging during reorienting tasks – such as the posner-cueing paradigm (Corbetta & Shulman, 2002; Konrad et al., 2005; Mitchell, 2008) or oddball detection tasks (Downar et al., 2001; Fichtenholtz et al., 2004; Stevens et al., 2000) – demonstrating greater activation in the TPJ when attentional reorienting or attentional detection was required. Neuropsychologists bolstered the argument, noting that spatial neglect was commonly associated with TPJ damage. One example is the claim that damage to the right TPJ is associated with attentional deficits in the contralateral visual field (Friedrich et al., 1998). However, this was later shown to be a mislocalization, with the crucial region being slightly more anterior (Smith et al., 2013).

Memory — specifically episodic memory retrieval — was also thought to be a primary cognitive function associated with the TPJ. Tasks that involved old/new recognition recruited the TPJ, primarily for successfully-retrieved old and successfully-rejected new words (Konishi et al., 2000). Confidence in episodic retrieval also correlated with TPJ activation, where greater activation was observed for responses with higher confidence (Cabeza et al., 2008; Kim & Cabeza, 2007; Wagner et al., 2005).

Others have argued that the TPJ was a purely social region, based on neuroimaging results with theory of mind tasks (i.e., inferring mental state of other social agents; Kestemont et al., 2013; Saxe & Powell, 2006) in a variety of formats: theory of mind stories (Aichhorn et al., 2009; Döhnelt et al., 2012; Kobayashi et al., 2007; Saxe & Powell, 2006; R. Saxe & Kanwisher, 2003; Young et al., 2007); inferring mental states from pictures (Rabin et al., 2010; Samson et al., 2008); eye gaze interpretation (Baron-Cohen et al., 1999; Mitchell et al., 2005; Platek et al.,

2004); and playing games with another social opponent (Gallagher et al., 2002; Kircher et al., 2009; Powell et al., 2017). Distinct patterns in the right TPJ were also associated with autism, a clinical condition linked with social communication deficits, suggesting that the TPJ is a region involved in social processing (Lombardo et al., 2011). Transcranial magnetic stimulation (TMS) studies also demonstrated that temporally disrupting the right TPJ affects reasoning about others' mental states (Young et al., 2010).

These studies have only served to fuel debate over function in the TPJ. With all of the conflicting information, researchers began to question the single-module view of the TPJ, which posited that the TPJ is entirely devoted to one cognitive process. In fact, studies that included direct contrasts of attention and social tasks found that parts of the TPJ became active during both tasks, but that there were also subregions of the TPJ that seemed to be more responsive to one function than the other (Mitchell, 2008).

Hence, following the debate over a singular function for the TPJ, researchers proposed that the TPJ is fractionated, i.e. that the region is a combination of single-function subunits. There are three lines of evidence supporting a fractionation view. First, within-study comparisons have found stronger activation in subregions of the TPJ for specific cognitive tasks (Hutchinson et al., 2014; Scholz et al., 2009). For example, Scholz and colleagues (2009) found neighboring but distinct peak activation (6-10mm spatial difference) for theory of mind and attentional reorienting tasks within participants. Second, anatomical heterogeneity has been identified within the TPJ; four subregions were extracted with task-free fMRI data using independent components analysis in the TPJ (Igelström et al., 2015). Lastly, meta-analyses have found that different cognitive processes are localized in different parts of the TPJ. For example, three spatially distinct regions were identified for three different tasks: theory of mind,

reorienting, and odd-ball detection (Kubit & Jack, 2013). Similarly, an independent component analysis with five different cognitive tasks – memory, social attention, reorienting, oddball detection, and theory of mind tasks – found five distinct activation profiles (Igelström et al., 2016), which supports the view of TPJ being comprised of multiple sub-regions.

In contrast to the fractionation view, others have argued for an “overarching view” of TPJ function (Cabeza et al., 2011). Such views hold that the TPJ is engaged in a single function that serves as an elemental ingredient of many tasks, with all contributing areas serving that function. For example, Geng and Vossel (2013) proposed that the TPJ enables "contextual updating", which is a combination of post-perceptual processes and top-down modulation. Others have proposed an overarching function of attention (Lückmann et al., 2014) or an overarching cognitive mechanism in attentional control and mentalizing (Schuwerk et al., 2017). This coordinated and holistic view of TPJ function is supported by a meta-analysis by Schurz et al. (Schurz et al., 2014) which finds areas of high consistency across a wide variety of social cognition tasks. However, neither the fractionation nor the overarching view addresses the fact that unique cognitive processes are supported in the TPJ and other higher-order regions (Carter et al., 2012).

In fact, the nexus model of TPJ function proposes that new and unique cognitive processes emerge as primary processes overlap, instead of being composed of a single function (Carter & Huettel, 2013). This combination produces a context for social predictions based on past experiences, and eventually leads to the emergence of a novel function that would not be possible without input from surrounding areas. This model, in other words, posits that the TPJ is not a set of independent sub-regions, as hypothesized in the fractionation view, but carries out a single novel and multifaceted function with processing characteristics of all of the surrounding

processes. Moreover, in contrast to the overarching view, the nexus model hypothesizes that the TPJ does not support any single cognitive domain, but instead combines information from each of these domains into a novel function.

3. Novel functions arise in nexus regions in the TPJ

There is, in fact, accumulating evidence that new composite functions are present in higher-order cognitive regions like the TPJ. Interestingly, support for convergence of functions comes from past studies originally meant to argue that the TPJ was a series of fractionated sub-modules. However, a different question could be asked: is there evidence of overlap amongst different cognitive functions? There is indeed clear evidence of overlap when studies combine paradigms from different areas of cognitive neuroscience (Fig. 2): memory and perception (Cabeza et al., 2011); theory of mind and attention (Krall et al., 2015; Mitchell, 2008); overlap of reorienting, empathy, and theory of mind (Decety & Lamm, 2007); and default-mode and social processes (Padmanabhan et al., 2017). With this substantial overlap of activation for distinct cognitive processes, it is important to ask if these areas of overlap are coincidental or evidence of interactions, producing novel cognitive processes. We review four papers characterizing cognitive processes in areas of overlapping activation within the TPJ.

Starting with one cognitive function, attention, Vossel and colleagues (2014) provide evidence of interaction between the two distinct networks (dorsal and ventral) within the TPJ. Early on, the TPJ was hypothesized to primarily serve as a bottom-up circuit breaker for on-going actions. However, recent studies have shown that responses in the TPJ are temporally delayed when compared to the dorsal attention network regions, such as the frontal eye fields and intraparietal sulcus, making it difficult to imagine that the TPJ is solely an early reorienting area.

Transcranial magnetic stimulation of the TPJ highlights its role as an interactive composite of the two attention systems without one explicitly directing or interrupting the other (Chica et al., 2011). In fact, the TPJ combines information from separate systems into unique representations in the brain, supporting the argument for interaction; for example, the TPJ integrates contextual information for non-target stimuli. Such activation was construed as attentional control signals of the TPJ, which are unique representations, formed only after interaction with the dorsal attentional regions take place (DiQuattro & Geng, 2011). Geng and Vossel (2013) conclude that ventral and dorsal attention networks develop separately, but by the time these networks converge in the TPJ, there is flexible interaction between the two systems that is dependent on the nature of the task. This suggests a role for convergence zones that is more than a simple extension of any one of the converging processing streams. It suggests convergence zones mediate interactions between disparate functions.

Evidence of new combinatorial functions also comes from research on attention and memory, which are traditionally thought to be independent. However, interactions between the two processes result in combined activation localized in convergence zones (Fig. 3b). The Attention-to-Memory model (Cabeza et al., 2008, 2011; Ciaramelli et al., 2008) posits that the dual attention network is also paralleled in memory processes, a circumstance which the convergence framework predicts would produce new combinations of these functions. In Cabeza's 2011 study, the superior parietal lobe performs strategic memory retrieval, a combination of memory and top-down attention from the dorsal streams. The inferior parietal lobe is activated by a combination of memory and bottom-up attention, where those streams converge. When these large scale gradients of attention, memory, and perception meet, they produce unique processes in predictable locations.

In addition, there is evidence of statistical interaction between social and attention processes. Schuwerk and colleagues (2017) designed an attention-orienting task (posner task) in a social vs non-social context to isolate the region (or regions) of the brain that support a task that requires interaction between processing streams. A univariate analysis revealed a significant interaction effect of attention and social conditions in the TPJ. In other words, the TPJ was differentially activated for a unique combination of attentional and social processes, not somewhat less active for either process separately.

Using a multivariate approach, the TPJ has also been found to be uniquely predictive of conditions across three domains of cognition: attention, memory, and social. Specifically, the TPJ involvement was in a setting when people were bluffing against a human opponent who was described as better, as opposed to when playing against a computer opponent (Carter et al., 2012; Fig. 3a). This indicates that a number of components are required for the TPJ to contribute predictable value in decision making: 1) only when the opponent is human (“social”), 2) while strategizing and considering one's past behavior (“memory”), and 3) only when the human opponent was considered better than the computer opponent (“worthy of attention”). The region did not provide predictive value when either the social or attention components were missing. This finding lends support to the argument that converging processing streams lead to novel complex processing, not possible if the region was fractionated into modular functions.

We have presented evidence that novel functions arise in nexus-like regions, as predicted by the nexus model. Now, we seek to explain how and where nexus regions occur. While the nexus model asks "what is a nexus area?", the convergence framework asks "how do nexus areas, like the TPJ, arise?". In other words, the nexus model focuses on the novel cognitive processes within a nexus area as a combination of multiple cognitive domains; here, we focus on

the mechanism — convergence of hierarchical abstracted gradients — that produces a nexus area.

4. How does convergence arise in the brain?

In this section, we describe three characteristics of convergent processing, using the TPJ for illustration. Macroscale gradients have origins in the sensory processing system and travel up to higher-order cognitive regions like the TPJ. Information within these gradients is hierarchical and becomes more abstract as it becomes distanced from sensory areas. As the gradients follow an axis of hierarchical abstraction and move away from primary cortices, a geographical space issue occurs: gradients run out of space. This is where convergent brain areas should occur, maximally distant from the primary sensory areas. Here, we describe evidence for three characteristics: 1) information is processed in hierarchies from primary sensory areas, 2) distance from primary sensory areas is correlated with representation abstraction, and 3) sensory processing streams terminate at points maximally distant from primary sensory cortices.

4.1. Information is processed in a hierarchical manner, from primary sensory areas to higher-order cognitive regions.

The human brain follows a semi-modular hierarchical organization (Bullmore & Sporns, 2009; Hubel & Wiesel, 1962; Meunier et al., 2009; Van Essen & Maunsell, 1983; Zhou et al., 2006), in which early neurons are simple and later neurons are more complex, clusters of neurons form functional modules, and these modules are hierarchically organized. We use a relatively well-established example of hierarchical organization in the brain, the visual cortex (Fig. 1a). For example, when we look at a human face, the primary visual cortex (V1) processes low-level

information, such as edges of a particular orientation, in a particular location. These representations are then fed to higher visual areas that connect neighboring edges into lines and curves (V2), textures (V3/V4), and so on, until a holistic face is represented (lateral occipital face area IOFA). This is a computationally reasonable strategy that is in line with models of computer vision (Szeliski, 2011). Hierarchical processing in the visual stream then takes on a less spatial characteristic as the calculation of motion becomes dominant. In social processing, this supports the detection of emotional expressions and eye movements (superior temporal sulcus; STS; Engell & Haxby, 2007; Narumoto et al., 2001; Zhang et al., 2020). The calculation of temporal predictions also serves as a mechanism for generating predictions about another's internal state and future actions (TPJ). What started out as simple lines and facial forms has transformed into novel, higher-order, abstract information. Other sensory processing streams, such as auditory (Kikuchi et al., 2010; Merzenich & Brugge, 1973; Wessinger et al., 2001) and somatomotor processing (Iwamura, 1998; Iwamura et al., 1993; Vaessen et al., 2019), also follow this pattern of hierarchical abstraction (Fig. 1b, 1c). These findings are also supported by neuronal recording in animals (Merzenich & Brugge, 1973) and neuroimaging in humans (Sepulcre et al., 2010). Having established the hierarchical organization of the brain, we can now turn to the content of information within this hierarchy.

4.2. The abstractness of a representation increases with distance from primary sensory areas.

Within these information processing hierarchies, we alluded to increasing complexity of representation. We will make that relationship more explicit in this section. As representations move further away from primary sensory cortices, they become domain-general, less detailed,

and are stripped of their physical features, resulting in higher-order representations, such as social intentions, memories of the past, or linguistic connotations. In fact, this notion was proposed early on by Mesulam (1998), who theorized that sensory inputs become abstracted as unimodal gradients converge towards the transmodal cortex. Building on Mesulam's theory, we hypothesize that the convergence of hierarchically abstracted gradients is the key reason we observe higher-order cognition reliably associated with convergence areas.

The role of abstraction in these hierarchical gradients is to deliver information in a biologically-efficient manner. Returning to the example of visualizing a face, what started out as minute details of edges and orientations in V1 are later combined to represent more complex features (Croner & Kaplan, 1995; Van Essen & Maunsell, 1983). Eventually, this combination of information is enough to represent a human face. However, combining information in raw form is expensive, due to storage space and energy cost. It is important to note that the metabolic costs and the information obtained is not a one-to-one relationship: increasing information rate two-fold requires more than doubling the energy costs (Balasubramanian & Sterling, 2009). Abstraction provides a more compact representation of information contained at lower levels, increasing the density of information and providing a simple means to combine complex representations. In addition to abstraction from area to area, there are strong indications of increasing abstraction across the brain as a whole. For example, when semantic representations are mapped onto the cortical surface (Huth et al., 2016), a gradient emerges beginning with concrete words of perceptual/physical attributes (e.g. "fingers" and "four") and extending to abstract words of social and emotional attributes (e.g. "child" and "despised"). This concrete-abstract gradient aligns with the same neural hierarchy of unimodal sensorimotor regions and transmodal higher-order areas described above.

Adding another line of evidence to the abstraction argument, Fernandino and colleagues (2016) mapped words of sensory-motor attributes (color, shape, manipulation, sound, and motion) and found clear dimensions of increasing abstraction (Fig. 2d). Each attribute was associated with activation in its corresponding sensory-motor region, whereas the conjunction of these attributes, such as color/sound/motion, were mapped primarily onto zones of convergence. These findings support the concrete-abstract gradient along the sensory hierarchical axis.

This localization of abstract representations in convergence zones is non-trivial. While it is generally accepted that abstraction increases in a posterior-to-anterior gradient (Taren et al., 2011), the wide distribution of abstract information to convergence zones found in a meta-analysis by Desai and colleagues (2018) was not expected. In fact, the convergence framework predicts that the distribution of abstract representation will occur in these areas maximally distant from primary sensory-motor regions, where macroscale gradients converge.

4.3. Convergence zones, the TPJ among them, are maximally and equidistant from primary sensory-motor areas.

Hierarchical abstraction cannot continue on forever in a limited brain space. We can therefore predict where convergence occurs. First, gradients converge at points equidistant from primary sensory cortices; any point that lies on the perpendicular bisector of, for example, the visual and auditory cortices will be equidistant from these cortices, following the properties of a triangle. Second, gradients also converge at a point maximally distant from primary sensory-motor cortices. In fact, Margulies and colleagues (2016) found a spatial topography where higher-order regions are not only maximally distant from sensory regions, but also equally distant from each

primary sensory cortex in terms of geodesic distance (Fig. 4a). This equidistance from sensory regions allows for higher-order regions to conveniently receive incoming sensory gradients.

Furthermore, this topography of maximal and equidistance aligns with the concrete-abstract gradients. Margulies and colleagues (2016) show that aligning topic term reverse-inference activation maps from neurosynth reveals precisely this type of hierarchical organization of abstraction. Again, using geodesic distance, primary sensory cortices and regions closer to the sensory origins were correlated with concrete terms such as motor and eye movement, whereas higher-order regions (maximally and equidistantly located from sensory origins) were correlated with complex and abstract terms such as emotion and social cognition. Furthermore, using both the equidistance maps from Margulies et al. (2016) and the gradient of abstract semantic representation from Huth et al. (2016), Huntenburg and colleagues (2018) demonstrate clear alignment between geodesic distance and concrete-abstract gradients. In other words, gradients that do converge will not only be furthest away from sensory cortices but will also have the most abstract, complex content. This further corroborates the idea that information in macroscale gradients becomes hierarchical and abstract as it moves away from sensory origins.

Being located on a maximal-equidistant point from primary sensory cortices and being situated at the apex of the concrete abstract gradient creates a setting for rich information to emerge, once the gradients do converge. With this cascade of information integration, by the time information reaches locations like the TPJ, the combined information will be optimally processed into a massive multidimensional space, resulting in unique, novel higher-order functions, which will allow people to flexibly navigate a complex world.

We have outlined evidence for a macroscale structure in the brain, consisting of processing streams of increasing abstraction and hierarchy. As the cortical area is limited, these streams will eventually run into each other. This presents an opportunity: computationally expensive representations can be combined. In accordance with this prediction, there is evidence to support the presence of novel functions that are a combination of the converging gradients (Cabeza et al., 2011; Carter et al., 2012; Schuwerk et al., 2017), as well as indications of some functional differentiation surrounding convergence zones that reflect neighboring cortical functions (Schurz et al., 2014). Convergence zone functions are not just an addition of two independent functions, but a novel combination of highly abstract representations, making convergence zones more than just the activation of surrounding gradients. Next, we discuss the evidence for evolutionary and developmental characteristics that are consistent with convergent processing streams.

5. Evolutionary and developmental properties of convergence zones

The progressive abstraction of representations in convergent processing streams is reminiscent of changes that occur over time in evolution and development. Evolutionarily, convergence zones are expanded in humans compared to other species (Hill et al., 2010; Fig. 4e). This could at least partially explain the enhanced role that cognitive functions inferring another's mental state play in human behavior. Developmentally, the function of convergence zones becomes apparent relatively late (Fig. 4d). Using vision as an example, infants tend to respond to high-contrast edges before more complex shapes, distinguishing biological movement by six months (Woodward, 1998). Continuing this progression, successful mental state inference occurs much later (3-5 years depending on the task used; Hughes & Leekam, 2004). As such, brain areas

further away from primary sensory cortices take longer to fully form. In fact, convergence zones, including the TPJ, undergo the largest cortical expansion after gestation (Fjell et al., 2015; Hill et al., 2010), continuing to develop until young adulthood. The late development of these regions mirrors their evolutionary expansion as well as their place in the computational hierarchy. This late development has additional side effects. Because they develop after a certain amount of world experience, they are less tied to genetic influence and more related to formative experiences. As a result, they should be the most variable between individuals. In fact, the TPJ is one of the most heterogeneous areas in the brain (Croxson et al., 2018; Mueller et al., 2013; Van Essen & Dierker, 2007; Fig. 4b, 4c). This idea is in line with the tethering hypothesis suggested by Buckner and Krienen (Buckner & Krienen, 2013), which posits that associative cortices are regions with relatively fewer constraints in terms of development, supporting more variability. In sum, the convergence account provides a framework for understanding the development of the TPJ and other convergence zones, which ties together disparate findings that span from equidistance from sensory cortices, late development, heterogeneity across individuals, and expansion in humans from evolutionary relatives.

6. Conclusion

The TPJ is not the only zone of convergence. A number of regions, such as the ventral medial prefrontal cortex, anterior temporal lobe, inferior frontal gyrus, dorsal medial prefrontal cortex, and posterior cingulate cortex, share common characteristics of the predicted properties of convergence zones. They are generally identified as part of the default mode network, located equidistantly from primary convergence zones, developing late within an individual, and are heterogeneous across individuals. The convergence framework provides a unifying view for

predicting the location and function of these regions. Given knowledge of the contributing processing streams, these predictions could be used to design carefully targeted experiments that reveal novel, multifaceted functions in the brain.

Acknowledgements

We would like to thank Tim Curran and Elizabeth C. Tremmel for valuable comments on the manuscript.

This work was supported by the Institute of Cognitive Science at the University of Colorado Boulder. HJ was also supported by the National Institutes of Health [grant numbers EB026549-01, MH116026].

Appendix

NA

References

- Aichhorn, M., Perner, J., Weiss, B., Kronbichler, M., Staffen, W., & Ladurner, G. (2009). Temporo-parietal junction activity in theory-of-mind tasks: falseness, beliefs, or attention. *Journal of Cognitive Neuroscience*, 21(6), 1179–1192. <https://doi.org/10.1162/jocn.2009.21082>
- Balasubramanian, V., & Sterling, P. (2009). Receptive fields and functional architecture in the retina. *The Journal of Physiology*, 587(Pt 12), 2753–2767. <https://doi.org/10.1113/jphysiol.2009.170704>
- Barbas, H. (1986). Pattern in the laminar origin of corticocortical connections. *The Journal of Comparative Neurology*, 252(3), 415–422. <https://doi.org/10.1002/cne.902520310>
- Baron-Cohen, S., Ring, H. A., Wheelwright, S., Bullmore, E. T., Brammer, M. J., Simmons, A., & Williams, S. C. R. (1999). Social intelligence in the normal and autistic brain: an fMRI study. *The European Journal of Neuroscience*, 11(6), 1891–1898. <https://onlinelibrary.wiley.com/doi/abs/10.1046/j.1460-9568.1999.00621.x>
- Baumgaertner, A., Buccino, G., Lange, R., McNamara, A., & Binkofski, F. (2007). Polymodal conceptual processing of human biological actions in the left inferior frontal lobe: Conceptual processing of biological hand actions. *The European Journal of Neuroscience*, 25(3), 881–889. <https://doi.org/10.1111/j.1460-9568.2007.05346.x>
- Buckner, R. L., & Krienen, F. M. (2013). The evolution of distributed association networks in the human brain. *Trends in Cognitive Sciences*, 17(12), 648–665. <https://doi.org/10.1016/j.tics.2013.09.017>
- Bukowski, H., Lamm, C., Zeigler-Hill, V., & Shackelford, T. (2017). Temporoparietal junction. *Encyclopedia of Personality and Individual Differences*, 1–5. https://www.researchgate.net/profile/Henryk_Bukowski/publication/318917343_Temporoparietal_Junction/links/5ab24e49458515e4cebedcc9b/Temporoparietal-Junction.pdf
- Bullmore, E., & Sporns, O. (2009). Complex brain networks: graph theoretical analysis of structural and functional systems. *Nature Reviews. Neuroscience*, 10(3), 186–198. <https://doi.org/10.1038/nrn2575>

- Cabeza, R., Ciaramelli, E., Olson, I. R., & Moscovitch, M. (2008). The parietal cortex and episodic memory: an attentional account. *Nature Reviews. Neuroscience*, 9(8), 613–625.
<https://doi.org/10.1038/nrn2459>
- Cabeza, R., Mazuz, Y. S., Stokes, J., Kragel, J. E., Woldorff, M. G., Ciaramelli, E., Olson, I. R., & Moscovitch, M. (2011). Overlapping parietal activity in memory and perception: evidence for the attention to memory model. *Journal of Cognitive Neuroscience*, 23(11), 3209–3217.
https://doi.org/10.1162/jocn_a_00065
- Carter, R. M., Bowling, D. L., Reeck, C., & Huettel, S. A. (2012). A distinct role of the temporal-parietal junction in predicting socially guided decisions. *Science*, 337(6090), 109–111.
<https://doi.org/10.1126/science.1219681>
- Chica, A. B., Lasaponara, S., Chanes, L., Valero-Cabré, A., Doricchi, F., Lupiáñez, J., & Bartolomeo, P. (2011). Spatial attention and conscious perception: the role of endogenous and exogenous orienting. *Attention, Perception & Psychophysics*, 73(4), 1065–1081. <https://doi.org/10.3758/s13414-010-0082-6>
- Ciaramelli, E., Grady, C. L., & Moscovitch, M. (2008). Top-down and bottom-up attention to memory: a hypothesis (AtoM) on the role of the posterior parietal cortex in memory retrieval. *Neuropsychologia*, 46(7), 1828–1851. <https://doi.org/10.1016/j.neuropsychologia.2008.03.022>
- Corbetta, M., & Shulman, G. L. (2002). Control of goal-directed and stimulus-driven attention in the brain. *Nature Reviews. Neuroscience*, 3(3), 201–215. <https://doi.org/10.1038/nrn755>
- Croner, L. J., & Kaplan, E. (1995). Receptive fields of P and M ganglion cells across the primate retina. *Vision Research*, 35(1), 7–24. [https://doi.org/10.1016/0042-6989\(94\)e0066-t](https://doi.org/10.1016/0042-6989(94)e0066-t)
- Croxson, P. L., Forkel, S. J., Cerliani, L., & Thiebaut de Schotten, M. (2018). Structural variability across the primate brain: a cross-species comparison. *Cerebral Cortex*, 28(11), 3829–3841.
<https://academic.oup.com/cercor/article-abstract/28/11/3829/4508772>
- Damasio, A. R. (1989). The Brain Binds Entities and Events by Multiregional Activation from Convergence Zones. *Neural Computation*, 1(1), 123–132. <https://doi.org/10.1162/neco.1989.1.1.123>

- Decety, J., & Lamm, C. (2007). The role of the right temporoparietal junction in social interaction: how low-level computational processes contribute to meta-cognition. *The Neuroscientist: A Review Journal Bringing Neurobiology, Neurology and Psychiatry*, 13(6), 580–593.
<https://doi.org/10.1177/1073858407304654>
- Desai, R. H., Reilly, M., & van Dam, W. (2018). The multifaceted abstract brain. *Philosophical Transactions of the Royal Society of London. Series B, Biological Sciences*, 373(1752).
<https://doi.org/10.1098/rstb.2017.0122>
- DiQuattro, N. E., & Geng, J. J. (2011). Contextual knowledge configures attentional control networks. *The Journal of Neuroscience: The Official Journal of the Society for Neuroscience*, 31(49), 18026–18035. <https://doi.org/10.1523/JNEUROSCI.4040-11.2011>
- Döhl, K., Schuwerk, T., Meinhardt, J., Sodian, B., Hajak, G., & Sommer, M. (2012). Functional activity of the right temporo-parietal junction and of the medial prefrontal cortex associated with true and false belief reasoning. *NeuroImage*, 60(3), 1652–1661.
<https://doi.org/10.1016/j.neuroimage.2012.01.073>
- Downar, J., Crawley, A. P., Mikulis, D. J., & Davis, K. D. (2001). The effect of task relevance on the cortical response to changes in visual and auditory stimuli: an event-related fMRI study. *NeuroImage*, 14(6), 1256–1267. <https://doi.org/10.1006/nimg.2001.0946>
- Engell, A. D., & Haxby, J. V. (2007). Facial expression and gaze-direction in human superior temporal sulcus. *Neuropsychologia*, 45(14), 3234–3241.
<https://doi.org/10.1016/j.neuropsychologia.2007.06.022>
- Fairhall, S. L., & Caramazza, A. (2013). Brain regions that represent amodal conceptual knowledge. *The Journal of Neuroscience: The Official Journal of the Society for Neuroscience*, 33(25), 10552–10558. <https://doi.org/10.1523/JNEUROSCI.0051-13.2013>
- Felleman, D. J., & Van Essen, D. C. (1991). Distributed hierarchical processing in the primate cerebral cortex. *Cerebral Cortex*, 1(1), 1–47. <https://doi.org/10.1093/cercor/1.1.1>
- Fernandino, L., Binder, J. R., Desai, R. H., Pendl, S. L., Humphries, C. J., Gross, W. L., Conant, L. L., &

- Seidenberg, M. S. (2016). Concept Representation Reflects Multimodal Abstraction: A Framework for Embodied Semantics. *Cerebral Cortex*, 26(5), 2018–2034.
<https://doi.org/10.1093/cercor/bhv020>
- Fichtenholtz, H. M., Dean, H. L., Dillon, D. G., Yamasaki, H., McCarthy, G., & LaBar, K. S. (2004). Emotion–attention network interactions during a visual oddball task. *Cognitive Brain Research*, 20(1), 67–80. <https://doi.org/10.1016/j.cogbrainres.2004.01.006>
- Fjell, A. M., Grydeland, H., Krogsrud, S. K., Amlien, I., Rohani, D. A., Ferschmann, L., Storsve, A. B., Tamnes, C. K., Sala-Llloch, R., Due-Tønnessen, P., Bjørnerud, A., Sølsnes, A. E., Håberg, A. K., Skranes, J., Bartsch, H., Chen, C.-H., Thompson, W. K., Panizzon, M. S., Kremen, W. S., ... Walhovd, K. B. (2015). Development and aging of cortical thickness correspond to genetic organization patterns. *Proceedings of the National Academy of Sciences of the United States of America*, 112(50), 15462–15467. <https://doi.org/10.1073/pnas.1508831112>
- Friedrich, F. J., Egly, R., Rafal, R. D., & Beck, D. (1998). Spatial attention deficits in humans: a comparison of superior parietal and temporal-parietal junction lesions. *Neuropsychology*, 12(2), 193–207. <https://doi.org/10.1037//0894-4105.12.2.193>
- Gallagher, H. L., Jack, A. I., Roepstorff, A., & Frith, C. D. (2002). Imaging the intentional stance in a competitive game. *NeuroImage*, 16(3 Pt 1), 814–821. <https://doi.org/10.1006/nimg.2002.1117>
- Geng, J. J., & Vossel, S. (2013). Re-evaluating the role of TPJ in attentional control: contextual updating? *Neuroscience and Biobehavioral Reviews*, 37(10 Pt 2), 2608–2620.
<https://doi.org/10.1016/j.neubiorev.2013.08.010>
- Hill, J., Inder, T., Neil, J., Dierker, D., Harwell, J., & Van Essen, D. (2010). Similar patterns of cortical expansion during human development and evolution. *Proceedings of the National Academy of Sciences of the United States of America*, 107(29), 13135–13140.
<https://doi.org/10.1073/pnas.1001229107>
- Hubel, D. H., & Wiesel, T. N. (1962). Receptive fields, binocular interaction and functional architecture in the cat's visual cortex. *The Journal of Physiology*, 160, 106–154.

<https://www.ncbi.nlm.nih.gov/pubmed/14449617>

- Hughes, C., & Leekam, S. (2004). What are the Links Between Theory of Mind and Social Relations? Review, Reflections and New Directions for Studies of Typical and Atypical Development. *Social Development*, 13(4), 590–619. <https://doi.org/10.1111/j.1467-9507.2004.00285.x>
- Huntenburg, J. M., Bazin, P.-L., & Margulies, D. S. (2018). Large-Scale Gradients in Human Cortical Organization. *Trends in Cognitive Sciences*, 22(1), 21–31. <https://doi.org/10.1016/j.tics.2017.11.002>
- Hutchinson, J. B., Uncapher, M. R., Weiner, K. S., Bressler, D. W., Silver, M. A., Preston, A. R., & Wagner, A. D. (2014). Functional heterogeneity in posterior parietal cortex across attention and episodic memory retrieval. *Cerebral Cortex*, 24(1), 49–66. <https://doi.org/10.1093/cercor/bhs278>
- Huth, A. G., de Heer, W. A., Griffiths, T. L., Theunissen, F. E., & Gallant, J. L. (2016). Natural speech reveals the semantic maps that tile human cerebral cortex. *Nature*, 532(7600), 453–458. <https://doi.org/10.1038/nature17637>
- Igelström, K. M., Webb, T. W., & Graziano, M. S. A. (2015). Neural Processes in the Human Temporoparietal Cortex Separated by Localized Independent Component Analysis. *The Journal of Neuroscience: The Official Journal of the Society for Neuroscience*, 35(25), 9432–9445. <https://doi.org/10.1523/JNEUROSCI.0551-15.2015>
- Igelström, K. M., Webb, T. W., Kelly, Y. T., & Graziano, M. S. A. (2016). Topographical Organization of Attentional, Social, and Memory Processes in the Human Temporoparietal Cortex. *eNeuro*, 3(2). <https://doi.org/10.1523/ENEURO.0060-16.2016>
- Iwamura, Y. (1998). Hierarchical somatosensory processing. *Current Opinion in Neurobiology*, 8(4), 522–528. [https://doi.org/10.1016/s0959-4388\(98\)80041-x](https://doi.org/10.1016/s0959-4388(98)80041-x)
- Iwamura, Y., Tanaka, M., Sakamoto, M., & Hikosaka, O. (1993). Rostrocaudal gradients in the neuronal receptive field complexity in the finger region of the alert monkey's postcentral gyrus. *Experimental Brain Research. Experimentelle Hirnforschung. Experimentation Cerebrale*, 92(3), 360–368. <https://doi.org/10.1007/BF00229023>
- Kestemont, J., Vandekerckhove, M., Ma, N., Van Hoeck, N., & Van Overwalle, F. (2013). Situation and

- person attributions under spontaneous and intentional instructions: an fMRI study. *Social Cognitive and Affective Neuroscience*, 8(5), 481–493. <https://doi.org/10.1093/scan/nss022>
- Kikuchi, Y., Horwitz, B., & Mishkin, M. (2010). Hierarchical auditory processing directed rostrally along the monkey's supratemporal plane. *The Journal of Neuroscience: The Official Journal of the Society for Neuroscience*, 30(39), 13021–13030. <https://doi.org/10.1523/JNEUROSCI.2267-10.2010>
- Kim, H., & Cabeza, R. (2007). Trusting our memories: dissociating the neural correlates of confidence in veridical versus illusory memories. *The Journal of Neuroscience: The Official Journal of the Society for Neuroscience*, 27(45), 12190–12197. <https://doi.org/10.1523/JNEUROSCI.3408-07.2007>
- Kircher, T., Blümel, I., Marjoram, D., Lataster, T., Krabbendam, L., Weber, J., van Os, J., & Krach, S. (2009). Online mentalising investigated with functional MRI. *Neuroscience Letters*, 454(3), 176–181. <https://doi.org/10.1016/j.neulet.2009.03.026>
- Kobayashi, C., Glover, G. H., & Temple, E. (2007). Children's and adults' neural bases of verbal and nonverbal “theory of mind.” *Neuropsychologia*, 45(7), 1522–1532. <https://doi.org/10.1016/j.neuropsychologia.2006.11.017>
- Konishi, S., Wheeler, M. E., Donaldson, D. I., & Buckner, R. L. (2000). Neural correlates of episodic retrieval success. *NeuroImage*, 12(3), 276–286. <https://doi.org/10.1006/nimg.2000.0614>
- Konrad, K., Neufang, S., Thiel, C. M., Specht, K., Hanisch, C., Fan, J., Herpertz-Dahlmann, B., & Fink, G. R. (2005). Development of attentional networks: an fMRI study with children and adults. *NeuroImage*, 28(2), 429–439. <https://doi.org/10.1016/j.neuroimage.2005.06.065>
- Krall, S. C., Rottschy, C., Oberwelland, E., Bzdok, D., Fox, P. T., Eickhoff, S. B., Fink, G. R., & Konrad, K. (2015). The role of the right temporoparietal junction in attention and social interaction as revealed by ALE meta-analysis. *Brain Structure & Function*, 220(2), 587–604. <https://doi.org/10.1007/s00429-014-0803-z>
- Kubit, B., & Jack, A. I. (2013). Rethinking the role of the rTPJ in attention and social cognition in light of the opposing domains hypothesis: findings from an ALE-based meta-analysis and resting-state functional connectivity. *Frontiers in Human Neuroscience*, 7, 323.

<https://doi.org/10.3389/fnhum.2013.00323>

Larivière, S., Vos de Wael, R., Hong, S.-J., Paquola, C., Tavakol, S., Lowe, A. J., Schrader, D. V., & Bernhardt, B. C. (2020). Multiscale Structure-Function Gradients in the Neonatal Connectome.

Cerebral Cortex, 30(1), 47–58. <https://doi.org/10.1093/cercor/bhz069>

Lombardo, M. V., Chakrabarti, B., Bullmore, E. T., MRC AIMS Consortium, & Baron-Cohen, S. (2011).

Specialization of right temporo-parietal junction for mentalizing and its relation to social impairments in autism. *NeuroImage*, 56(3), 1832–1838.

<https://doi.org/10.1016/j.neuroimage.2011.02.067>

Lückmann, H. C., Jacobs, H. I. L., & Sack, A. T. (2014). The cross-functional role of frontoparietal regions in cognition: internal attention as the overarching mechanism. *Progress in Neurobiology*, 116, 66–86. <https://doi.org/10.1016/j.pneurobio.2014.02.002>

Man, K., Kaplan, J., Damasio, H., & Damasio, A. (2013). Neural convergence and divergence in the mammalian cerebral cortex: from experimental neuroanatomy to functional neuroimaging. *The Journal of Comparative Neurology*, 521(18), 4097–4111.

<https://onlinelibrary.wiley.com/doi/abs/10.1002/cne.23408>

Margulies, D. S., Ghosh, S. S., Goulas, A., Falkiewicz, M., Huntenburg, J. M., Langs, G., Bezgin, G., Eickhoff, S. B., Castellanos, F. X., Petrides, M., Jefferies, E., & Smallwood, J. (2016). Situating the default-mode network along a principal gradient of macroscale cortical organization. *Proceedings of the National Academy of Sciences of the United States of America*, 113(44), 12574–12579.

<https://doi.org/10.1073/pnas.1608282113>

Merzenich, M. M., & Brugge, J. F. (1973). Representation of the cochlear partition of the superior temporal plane of the macaque monkey. *Brain Research*, 50(2), 275–296.

[https://doi.org/10.1016/0006-8993\(73\)90731-2](https://doi.org/10.1016/0006-8993(73)90731-2)

Mesulam, M. M. (1998). From sensation to cognition. *Brain: A Journal of Neurology*, 121 (Pt 6), 1013–1052. <https://doi.org/10.1093/brain/121.6.1013>

Meunier, D., Lambiotte, R., Fornito, A., Ersche, K. D., & Bullmore, E. T. (2009). Hierarchical modularity

- in human brain functional networks. *Frontiers in Neuroinformatics*, 3, 37.
<https://doi.org/10.3389/neuro.11.037.2009>
- Mitchell, J. P. (2008). Activity in right temporo-parietal junction is not selective for theory-of-mind. *Cerebral Cortex*, 18(2), 262–271. <https://doi.org/10.1093/cercor/bhm051>
- Mitchell, J. P., Banaji, M. R., & Macrae, C. N. (2005). The link between social cognition and self-referential thought in the medial prefrontal cortex. *Journal of Cognitive Neuroscience*, 17(8), 1306–1315. <https://doi.org/10.1162/0898929055002418>
- Mueller, S., Wang, D., Fox, M. D., Yeo, B. T. T., Sepulcre, J., Sabuncu, M. R., Shafee, R., Lu, J., & Liu, H. (2013). Individual variability in functional connectivity architecture of the human brain. *Neuron*, 77(3), 586–595. <https://doi.org/10.1016/j.neuron.2012.12.028>
- Narumoto, J., Okada, T., Sadato, N., Fukui, K., & Yonekura, Y. (2001). Attention to emotion modulates fMRI activity in human right superior temporal sulcus. *Brain Research. Cognitive Brain Research*, 12(2), 225–231. [https://doi.org/10.1016/s0926-6410\(01\)00053-2](https://doi.org/10.1016/s0926-6410(01)00053-2)
- Padmanabhan, A., Lynch, C. J., Schaer, M., & Menon, V. (2017). The Default Mode Network in Autism. *Biological Psychiatry. Cognitive Neuroscience and Neuroimaging*, 2(6), 476–486.
<https://doi.org/10.1016/j.bpsc.2017.04.004>
- Paquola, C., Vos De Wael, R., Wagstyl, K., Bethlehem, R. A. I., Hong, S.-J., Seidlitz, J., Bullmore, E. T., Evans, A. C., Misić, B., Margulies, D. S., Smallwood, J., & Bernhardt, B. C. (2019). Microstructural and functional gradients are increasingly dissociated in transmodal cortices. *PLoS Biology*, 17(5), e3000284. <https://doi.org/10.1371/journal.pbio.3000284>
- Platek, S. M., Keenan, J. P., Gallup, G. G., Jr, & Mohamed, F. B. (2004). Where am I? The neurological correlates of self and other. *Cognitive Brain Research*, 19(2), 114–122.
<https://www.sciencedirect.com/science/article/pii/S0926641003002921>
- Powell, J. L., Grossi, D., Corcoran, R., Gobet, F., & García-Fiñana, M. (2017). The neural correlates of theory of mind and their role during empathy and the game of chess: A functional magnetic resonance imaging study. *Neuroscience*, 355, 149–160.

<https://doi.org/10.1016/j.neuroscience.2017.04.042>

Rabin, J. S., Gilboa, A., Stuss, D. T., Mar, R. A., & Rosenbaum, R. S. (2010). Common and unique neural correlates of autobiographical memory and theory of mind. *Journal of Cognitive Neuroscience*, 22(6), 1095–1111.

<https://www.mitpressjournals.org/doi/abs/10.1162/jocn.2009.21344>

Samson, A. C., Zysset, S., & Huber, O. (2008). Cognitive humor processing: Different logical mechanisms in nonverbal cartoons—an fMRI study. *Social Neuroscience*, 3(2), 125–140.

<https://doi.org/10.1080/17470910701745858>

Saxe, R., & Kanwisher, N. (2003). People thinking about thinking people: The role of the temporo-parietal junction in “theory of mind.” *NeuroImage*, 19(4), 1835–1842.

[https://doi.org/10.1016/S1053-8119\(03\)00230-1](https://doi.org/10.1016/S1053-8119(03)00230-1)

Saxe, R., & Powell, L. J. (2006). It’s the thought that counts: specific brain regions for one component of theory of mind. *Psychological Science*, 17(8), 692–699. <https://doi.org/10.1111/j.1467-9280.2006.01768.x>

Scholz, J., Triantafyllou, C., Whitfield-Gabrieli, S., Brown, E. N., & Saxe, R. (2009). Distinct regions of right temporo-parietal junction are selective for theory of mind and exogenous attention. *PloS One*, 4(3), e4869. <https://doi.org/10.1371/journal.pone.0004869>

Schurz, M., Radua, J., Aichhorn, M., Richlan, F., & Perner, J. (2014). Fractionating theory of mind: a meta-analysis of functional brain imaging studies. *Neuroscience and Biobehavioral Reviews*, 42, 9–34. <https://doi.org/10.1016/j.neubiorev.2014.01.009>

Schuwerk, T., Schurz, M., Müller, F., Rupprecht, R., & Sommer, M. (2017). The rTPJ’s overarching cognitive function in networks for attention and theory of mind. *Social Cognitive and Affective Neuroscience*, 12(1), 157–168. <https://academic.oup.com/scan/article-abstract/12/1/157/2823716>

Sepulcre, J., Liu, H., Talukdar, T., Martincorena, I., Yeo, B. T. T., & Buckner, R. L. (2010). The organization of local and distant functional connectivity in the human brain. *PLoS Computational Biology*, 6(6), e1000808. <https://doi.org/10.1371/journal.pcbi.1000808>

- Smith, D. V., Clithero, J. A., Rorden, C., & Karnath, H.-O. (2013). Decoding the anatomical network of spatial attention. *Proceedings of the National Academy of Sciences of the United States of America*, 110(4), 1518–1523. <https://doi.org/10.1073/pnas.1210126110>
- Stevens, A. A., Skudlarski, P., Gatenby, J. C., & Gore, J. C. (2000). Event-related fMRI of auditory and visual oddball tasks. *Magnetic Resonance Imaging*, 18(5), 495–502. [https://doi.org/10.1016/s0730-725x\(00\)00128-4](https://doi.org/10.1016/s0730-725x(00)00128-4)
- Szeliski, R. (2011). *Computer Vision. Texts in Computer Science*. Springer London, London.
- Tamber-Rosenau, B. J., Dux, P. E., Tombu, M. N., Asplund, C. L., & Marois, R. (2013). Amodal processing in human prefrontal cortex. *The Journal of Neuroscience: The Official Journal of the Society for Neuroscience*, 33(28), 11573–11587. <https://doi.org/10.1523/JNEUROSCI.4601-12.2013>
- Taren, A. A., Venkatraman, V., & Huettel, S. A. (2011). A parallel functional topography between medial and lateral prefrontal cortex: evidence and implications for cognitive control. *The Journal of Neuroscience: The Official Journal of the Society for Neuroscience*, 31(13), 5026–5031. <https://doi.org/10.1523/JNEUROSCI.5762-10.2011>
- Vaessen, M. J., Abassi, E., Mancini, M., Camurri, A., & de Gelder, B. (2019). Computational feature analysis of body movements reveals hierarchical brain organization. *Cerebral Cortex*, 29(8), 3551–3560. <https://academic.oup.com/cercor/article-abstract/29/8/3551/5112932>
- Van Essen, D. C., & Dierker, D. L. (2007). Surface-based and probabilistic atlases of primate cerebral cortex. *Neuron*, 56(2), 209–225. <https://doi.org/10.1016/j.neuron.2007.10.015>
- Van Essen, D. C., & Maunsell, J. H. R. (1983). Hierarchical organization and functional streams in the visual cortex. *Trends in Neurosciences*, 6, 370–375. [https://doi.org/10.1016/0166-2236\(83\)90167-4](https://doi.org/10.1016/0166-2236(83)90167-4)
- Vázquez-Rodríguez, B., Suárez, L. E., Markello, R. D., Shafiei, G., Paquola, C., Hagmann, P., van den Heuvel, M. P., Bernhardt, B. C., Spreng, R. N., & Misic, B. (2019). Gradients of structure-function tethering across neocortex. *Proceedings of the National Academy of Sciences of the United States of America*, 116(42), 21219–21227. <https://doi.org/10.1073/pnas.1903403116>
- Vossel, S., Geng, J. J., & Fink, G. R. (2014). Dorsal and ventral attention systems: distinct neural circuits

- but collaborative roles. *The Neuroscientist: A Review Journal Bringing Neurobiology, Neurology and Psychiatry*, 20(2), 150–159. <https://doi.org/10.1177/1073858413494269>
- Wagner, A. D., Shannon, B. J., Kahn, I., & Buckner, R. L. (2005). Parietal lobe contributions to episodic memory retrieval. *Trends in Cognitive Sciences*, 9(9), 445–453. <https://doi.org/10.1016/j.tics.2005.07.001>
- Wagstyl, K., Ronan, L., Goodyer, I. M., & Fletcher, P. C. (2015). Cortical thickness gradients in structural hierarchies. *NeuroImage*, 111, 241–250. <https://doi.org/10.1016/j.neuroimage.2015.02.036>
- Wessinger, C. M., VanMeter, J., Tian, B., Van Lare, J., Pekar, J., & Rauschecker, J. P. (2001). Hierarchical organization of the human auditory cortex revealed by functional magnetic resonance imaging. *Journal of Cognitive Neuroscience*, 13(1), 1–7. <https://doi.org/10.1162/089892901564108>
- Woodward, A. L. (1998). Infants selectively encode the goal object of an actor's reach. *Cognition*, 69(1), 1–34. [https://doi.org/10.1016/s0010-0277\(98\)00058-4](https://doi.org/10.1016/s0010-0277(98)00058-4)
- Young, L., Cushman, F., Hauser, M., & Saxe, R. (2007). The neural basis of the interaction between theory of mind and moral judgment. *Proceedings of the National Academy of Sciences of the United States of America*, 104(20), 8235–8240. <https://doi.org/10.1073/pnas.0701408104>
- Young, L., Dodell-Feder, D., & Saxe, R. (2010). What gets the attention of the temporo-parietal junction? An fMRI investigation of attention and theory of mind. *Neuropsychologia*, 48(9), 2658–2664. <https://doi.org/10.1016/j.neuropsychologia.2010.05.012>
- Zhang, H., Japee, S., Stacy, A., Flessert, M., & Ungerleider, L. G. (2020). Anterior superior temporal sulcus is specialized for non-rigid facial motion in both monkeys and humans. *NeuroImage*, 116878. <https://doi.org/10.1016/j.neuroimage.2020.116878>
- Zhou, C., Zemanová, L., Zamora, G., Hilgetag, C. C., & Kurths, J. (2006). Hierarchical organization unveiled by functional connectivity in complex brain networks. *Physical Review Letters*, 97(23), 238103. <https://doi.org/10.1103/PhysRevLett.97.238103>

Figure Captions

Fig 1. Cortical hierarchies parallel gradients of abstraction in the brain. (a) Cortical hierarchies for processing visual, auditory and somatosensory modalities are hierarchically organized, beginning with primary sensory (“red”) to higher-order regions (“yellow”; (Wagstyl et al., 2015). Their hierarchical numeric scales are based upon prior literature on structural hierarchy: visual (from 1 to 9; (Felleman & Van Essen, 1991), auditory (from 1 to 7; (Barbas, 1986), and somatosensory (from 1 to 5; Felleman & Van Essen, 1991). **(b)** Moving from primary sensory to higher-order areas, representations become more abstracted, as seen in this schematic of levels of abstraction. Primary sensory regions are associated with low-level features (i.e. edges, phonemes, and single finger joints) and smaller receptive fields. In contrast, higher-order regions are associated with abstract concepts (i.e. faces, paragraphs, and elbow/shoulder joints) and larger receptive fields. The abstracted levels in our schematic are based on prior literature: visual (Dicarlo, Zoccolan, & Rust, 2012), auditory (Kikuchi et al., 2010; Merzenich & Brugge, 1973; Wessinger et al., 2001), and somatosensory (Iwamura et al., 1993).

Fig 2. As a convergence zone, the TPJ shows overlapping activation of higher-order cognitive functions. Each color corresponds to the contrast mask within each study **(a)** reorienting, ToM, and empathy (Decety & Lamm, 2007). **(b)** perception and memory (Cabeza et al., 2011). **(c)** ToM and attention (Mitchell, 2008). **(d)** color, sound, motion, manipulation, shape of words (Fernandino et al., 2016). **(e)** empathy and attention (Decety & Lamm, 2007). **(f)** DMN and ToM (Padmanabhan et al., 2017). ToM, theory of mind; DMN, default mode network. (Overlap of the original figures has been recolored to white for visualization purposes.)

Fig 3. Unique functions emerge in the TPJ, a result of interactions of multiple cognitive processes. (a) The TPJ carries out a combination of functions (uniquely predictive of social opponents worthy of attention) rather than performing functions in isolation (either attention or social). In a study where participants were engaged in strategic gameplay, predictive information for each human and computer opponent was analyzed (Carter et al., 2012). Data points on the dotted-black midline indicate a brain region with equal predictive information for both human and computer opponents, notably, the TPJ is distant from the midline. This indicates that the predictive power is higher only when 1) the opponent is human, 2) the opponent is considered better, and 3) past trial histories are considered. **(b)** A combination of attentional and memory functions is found in the TPJ. Cabeza and colleagues (2011) conducted a 2x2 study of memory and perception tasks. In the memory task, participants would “orient” to the first word of memorized word pairs and “detect” the last word of the pair, whereas in the perception task, one would “orient” to the first consonant of letter streams and “detect” the vowel. The main effect of detecting/orienting, collapsed across perception and memory tasks, are plotted in the left panel. The right panel figure depicts activation of orienting (“right top”) and detection-related activity (“right bottom”) in each of the perception (“red”) and memory (“green”) tasks. In accordance with the convergence framework, each blob represents a unique combinatorial function consisting of what were previously thought of as separate processes.

Fig 4. The evolutionary and developmental properties of convergence zones. Convergence zones — including the TPJ, ventral medial prefrontal cortex, anterior temporal lobe, inferior frontal gyrus, dorsal medial prefrontal cortex — are: **(a)** equidistant from each other (Margulies et al., 2016), more variable in **(b)** white matter and **(c)** gray matter (Croxson et al., 2018), the most expanded **(d)** postnatally and **(e)** evolutionarily (Hill et al., 2010).

Figure 1

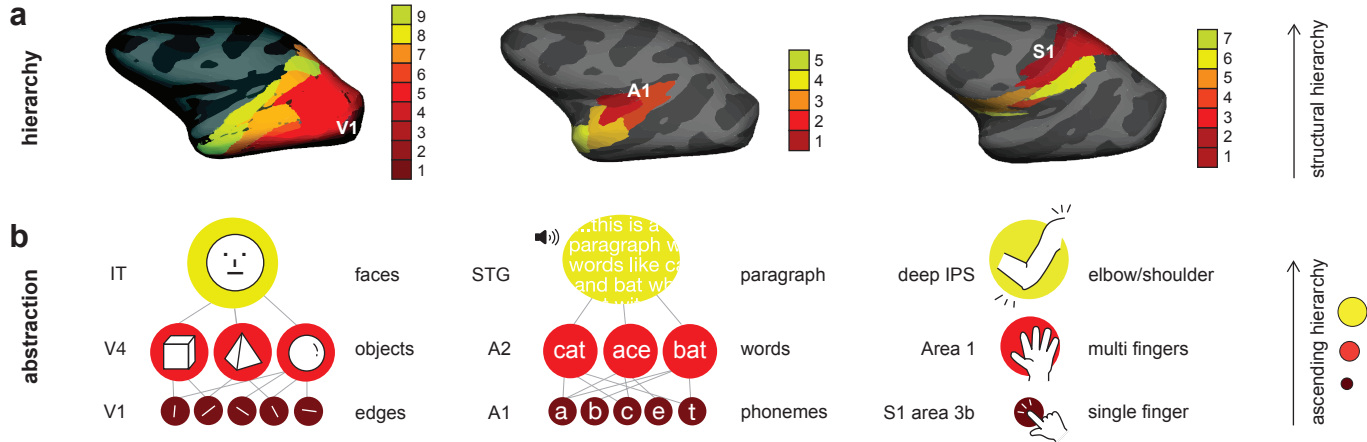


Figure 2

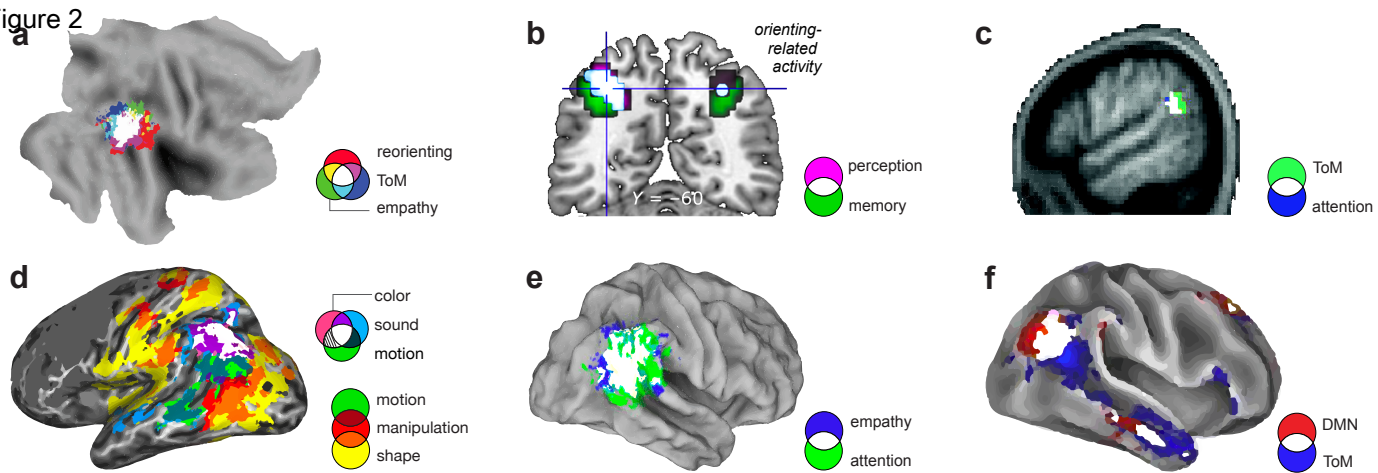
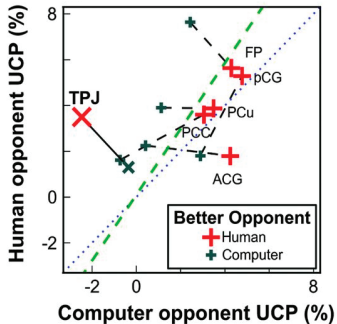


Figure 3



b

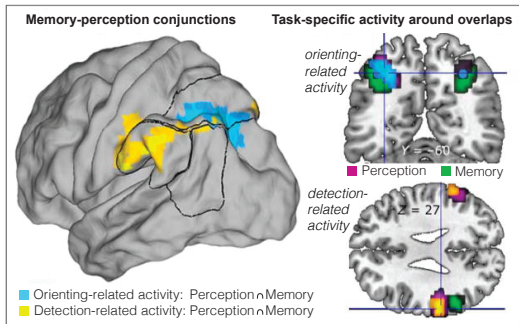
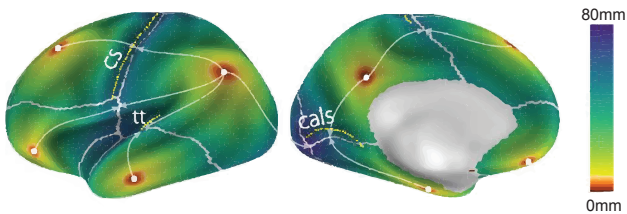
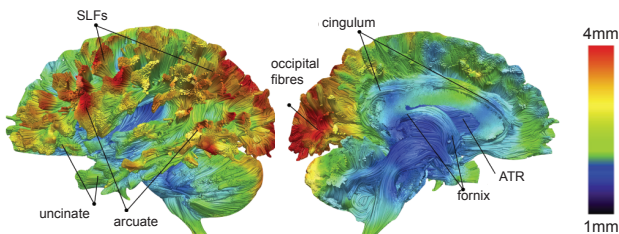


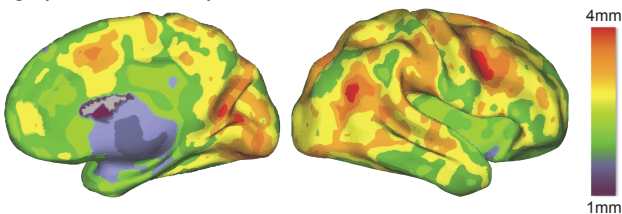
Figure 4
a equidistance



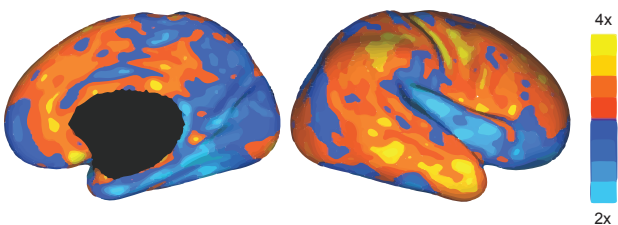
b white matter variability



c gray matter variability



d postnatal cortical expansion



e evolutionary cortical expansion

